

Reinis Cirpons

PhD Student

Education

- 2021/09-
Present **PhD in Mathematics**
 University Of St Andrews - St Andrews, UK
 • Expected to graduate 2025/12
- 2017/09-
2021/05 **Master of Mathematics (hons.) 1st class**
 University Of St Andrews - St Andrews, UK
 • Dean's List for academic excellence 2017-2021
 • President of the St Andrews Mathematics Society.

Publications

1. R. Cirpons, F. Hivert, A. Mahboubi, G. Melquiond, J. D. Mitchell, and F. Smith. Certifying the decidability of the word problem in monoids at large. In preparation., 2025a
2. R. Cirpons, J. D. Mitchell, and Y. Péresse. Maximal and minimal one sided congruences of the full transformation monoid. In preparation., 2025b
3. R. Cirpons, J. D. Mitchell, and F. L. Smith. Off with the head: termination provers and the word problem for 1-relation monoids. In *Proceedings of the 20th International Workshop on Termination*, September 2025c. URL https://www.imn.htwk-leipzig.de/WST2025/proceedings/WST2025_paper_16.pdf. To appear
4. R. Cirpons, J. East, and J. D. Mitchell. Transformation representations of diagram monoids, 2024. URL <https://arxiv.org/abs/2411.14693>
5. M. Anagnostopoulou-Merkouri, R. Cirpons, J. D. Mitchell, and M. Tsakalakou. Computing finite index congruences of finitely presented semigroups and monoids. *Math. Comp.*, 2025. URL <https://arxiv.org/abs/2302.06295>. To appear
6. R. Cirpons and J.D. Mitchell. Polynomial time multiplication and normal forms in free bands. *Theoretical Computer Science*, 953:113783, 2023. doi:[10.1016/j.tcs.2023.113783](https://doi.org/10.1016/j.tcs.2023.113783)

Invited talks

- **Off with the head: termination provers and the word problem for 1-relation monoids**,
20th International Workshop on Termination, Leipzig, Germany, September 2025.
- **Maximal one-sided congruences of full transformation monoids**,
Semigroups Day VI, St Andrews, UK, June 2025.
- **Computing finite-index congruences**,
75th British Mathematical Colloquium, Manchester, UK, June 2024.
- **Computing with free bands**,
York Semigroup seminar, York, UK, November 2023.
- **Making the Knuth-Bendix algorithm exponentially slower**,
Semigroups Day, St Andrews, UK, June 2023

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Links

Personal webpage

reinisc.id.lv

ORCID

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Github

[reinscirpons](https://github.com/reinscirpons)

Languages

English	
Latvian	
Russian	

Software contributions

- **The libsemigroups C++ library.**
A C++ library for computing with finite and finitely presented semigroups. Implemented finiteness criteria for monoids based on approaches from linear algebra, improved the implementation of the Sims algorithm for finite index congruences of monoids and helped with the changes for version 3 of the library.
URL: <https://github.com/libsemigroups/libsemigroups>
doi: [10.5281/zenodo.1437752](https://doi.org/10.5281/zenodo.1437752)
- **The freebandlib Python library.**
A Python library for computing with finitely generated free bands. Contains implementations of algorithms from “Polynomial time multiplication and normal forms in free bands”.
URL: <https://github.com/reiniscirpons/freebandlib>
doi: [10.5281/zenodo.7071675](https://doi.org/10.5281/zenodo.7071675)
- **The GAP system for computer algebra.**
A widely used open source computer algebra system. Improved orbit calculation code and extended the testing language.
URL: <https://github.com/gap-system/gap>
- **The Digraphs package for GAP.**
A GAP package implementing various algorithms computation with directed graphs. Implemented methods for computing the number of spanning trees of a graph, generating random lattices and finding maximum matchings. Improved orbit calculations for actions on digraphs via hashing.
URL: <https://github.com/digraphs/Digraphs>
doi: [10.5281/zenodo.596465](https://doi.org/10.5281/zenodo.596465)
- **The Semigroups package for GAP.**
A GAP package implementing various algorithms computation with semigroups. Improved computation with free bands.
URL: <https://github.com/semigroups/Semigroups>
doi: [10.5281/zenodo.592893](https://doi.org/10.5281/zenodo.592893)

Experience

2021/09- Present	Tutor <i>University Of St Andrews</i> - St Andrews, UK I tutor multiple small group tutorials each semester. I am also a lead demonstrator for the computing components of various modules.
2019/06- Present	Contributor <i>Digraphs</i> package and <i>libsemigroups</i> C++ library Have made many contributions to the Digraphs package and libsemigroups library as part of the team of contributors at the University of St Andrews. Gained valuable experience in Git, test driven development and learned a lot about the practical programming workflow.
2020/07- 2022/08	Junior Researcher <i>Whitecrypton</i> - Riga, LV Worked on whitebox cryptography and obfuscation. Analyzed the state of the art, prototyped possible solutions in Python, and found flaws in existing implementations to strengthen the security of the product.